

Unit 37: Integration by Parts

1. The idea, in plain words

Substitution (Unit 28) handles integrals built from the chain rule. Integration by parts is the tool for integrals built from the product rule — specifically, for integrals where two very different kinds of functions are multiplied together, like a polynomial times a trig function, or a polynomial times $\ln x$.

Where the formula comes from: recall the product rule, $\frac{d}{dx}[uv] = u'v + uv'$. Rearranging and integrating both sides eventually gives you:

$$\int u dv = uv - \int v du$$

Picture the whole idea like a trade. You're not directly integrating the original product — you're trading it for a different integral, $\int v du$, hoping that new integral is easier to handle than the one you started with. The entire skill of this unit is picking a good trade: choosing which factor becomes “ u ” (the piece you differentiate) and which becomes “ dv ” (the piece you integrate).

A simple guideline for choosing u — remember “LIATE”: given a choice, prefer to let u be whichever type of function appears earliest in this list: Logarithmic, Inverse trig, Algebraic (polynomials), Trig, Exponential. The idea is that u should be the piece that gets simpler when you differentiate it (like $\ln x$ turning into $\frac{1}{x}$, or x^2 turning into $2x$), while dv should be the piece that's easy to integrate no matter what (like e^x or $\cos x$, which don't get more complicated).

One more trick you'll need almost immediately: what if there's only one function, not two? Something like $\int \ln x dx$ or $\int \arctan x dx$ doesn't look like a product at all — there's no obvious second factor to split off. The move: invent one. Rewrite it as $\int 1 \cdot \ln x dx$, then let $u = \ln x$ and $dv = 1 dx$ (so $v = x$). Multiplying by an invisible “1” turns a lone function into a product you can attack normally — this exact move shows up constantly for standalone logs and inverse trig functions.

Sometimes you'll need to repeat the whole process if the new integral $\int v du$ is still a product needing its own round of integration by parts (this happens with things like $x^2 e^x$).

A genuinely surprising special case: “cyclic” integrals. For something like $\int e^x \cos x dx$, applying integration by parts twice actually brings back the original integral on the other side of the equation — at which point you solve for it algebraically, the same way you'd solve $I = 5 - I$ for I .

2. Toolbox

Integration by Parts formula:

$$\int u dv = uv - \int v du$$

The procedure:

1. Identify u and dv from the integrand (use LIATE as a guide for choosing u).
2. Compute du (the derivative of u) and v (the antiderivative of dv).
3. Plug into the formula: $uv - \int v du$.
4. Evaluate the new integral $\int v du$ — sometimes directly, sometimes with another substitution, and sometimes with a second round of integration by parts.

5. Add $+C$ at the very end (only once, on the final result).

For a “cyclic” integral (where the original integral reappears after two rounds): treat the whole equation algebraically — move all copies of the original integral to one side, factor, and divide to solve for it.

3. Common mistakes

- Choosing u and dv poorly, so that the new integral $\int v du$ is harder than the one you started with. If this happens, try swapping your choice of u and dv .
- Forgetting the minus sign in the formula. It's $uv - \int v du$, not $uv + \int v du$.
- Losing track of signs during a repeated or cyclic application. These problems often stack up several negative signs — go slowly and double-check each step.
- Stopping a cyclic integral partway through, expecting it to resolve normally. If the original integral reappears, that's the signal to solve for it algebraically, not a sign that something went wrong.
- Forgetting a constant factor when finding v — for instance, if $dv = \cos(3x) dx$, then $v = \frac{1}{3}\sin(3x)$, not just $\sin(3x)$.
- Adding $+C$ too early, or adding it more than once during a multi-step problem — it only belongs on the very final answer.

4. Problem Set

Warm-up

1. $\int x \cos x dx$
2. $\int x e^{2x} dx$
3. $\int x \sin(2x) dx$
4. $\int \ln x dx$
5. $\int (x+1)e^x dx$
6. $\int x e^{-x} dx$

Standard

7. $\int x \sin x dx$
8. $\int x^5 \ln x dx$
9. $\int x^2 e^x dx$ (this one needs two rounds!)
10. $\int \ln(x^2) dx$
11. $\int x \cos(3x) dx$
12. $\int \arctan(x) dx$

13. $\int x^3 \ln x \, dx$

Challenge

14. $\int x e^x \, dx$

15. $\int e^x \cos x \, dx$ (a cyclic integral!)

16. $\int e^{2x} \sin x \, dx$ (another cyclic integral)

17. $\int x^2 \sin x \, dx$ (two rounds needed)

18. $\int x(\ln x)^2 \, dx$ (two rounds needed)

Applied

19. A variable force is modeled by $F(x) = x \sin x$ (newtons). Find the total work done,
 $\int_0^\pi x \sin x \, dx$.

20. A continuous income stream has a present-value rate of te^{-t} . Find the total present value,
 $\int_0^{10} te^{-t} \, dt$.

21. Find $\int_1^e x \ln x \, dx$.

22. Find $\int_0^{\pi/2} x \cos x \, dx$.